

Product Optimization & Revenue Analysis

Afficionado Coffee Roasters

1. Introduction

In a coffee shop business, all products do not perform equally. Some products sell more, some generate higher revenue, and some have very little impact. This project focuses on understanding product performance and identifying key revenue drivers.

2. Problem Statement

Although transaction data is available, the business lacks clarity on profitable products, category contribution, and low-performing items. As a result, decisions are not fully data-driven.

3. Objectives

Primary: Identify top and least selling products, measure revenue contribution, and analyze revenue distribution. Secondary: Identify high-performing products and support better menu decisions.

4. Dataset Overview

The dataset includes transaction-level data such as quantity sold, unit price, product category, product type, and product details.

5. Methodology

Data was processed using Python (Pandas). Revenue was calculated as $\text{Quantity} \times \text{Price}$. Product and category level analysis along with Pareto analysis was performed.

6. Key Findings

Top Products: Earl Grey Rg, Dark Chocolate Lg, Latte. Tea and Coffee dominate sales.

Top Revenue Products: Sustainably Grown Organic Lg, Dark Chocolate Lg, Latte Rg. Premium products generate higher revenue.

Category Contribution: Coffee (38.63%), Tea (28.10%), Bakery (11.78%), Drinking Chocolate (10.36%). Coffee and Tea contribute ~66% revenue.

Low Performing Categories: Flavours, Packaged Chocolate, Loose Tea.

Pareto Insight: Top 10 products contribute around 25% revenue.

7. Recommendations

Focus on high revenue products, improve or remove low-performing products, strengthen coffee category, and introduce combo offers.

8. Conclusion

A small number of products drive most of the revenue. By focusing on high-performing products and optimizing the menu, the business can improve profitability.

9. Tools Used

Python (Pandas), Streamlit